

CSI 3140 - WWW Structures, Techniques and Standards**Laboratory 2****CSS Styling, Box Model, Responsive Design, and Flexbox/Grid Layout**

Professor	Mohamed Ali Ibrahim, PhD, PEng
Term	Spring/Summer 2026
Lab Period	Monday, June 1, 2026 to Saturday, June 13, 2026
Submission	Brightspace, by Saturday, June 13, 2026, 11:59 PM
Team Size	2-3 students
Technologies	HTML5, CSS3, media queries, Flexbox, CSS Grid, browser developer tools, W3C/CSS validation tools

1. Laboratory Overview

The purpose of this laboratory is to give students hands-on practice with modern CSS3 for styling, layout, spacing, responsive design, and visual organization. Students will extend or redesign a small multi-page website so that it demonstrates correct use of external stylesheets, the CSS box model, responsive layouts, Flexbox, and CSS Grid.

This laboratory is aligned with the course topics related to HTML/CSS integration, selectors, the cascade, inheritance, specificity, typography, color, spacing, the box model, responsive Web design, media queries, Flexbox, CSS Grid, browser developer tools, validation, and accessibility-aware visual design.

2. Topics Covered in Lab 2

Topic Area	Required Coverage
CSS foundations	External stylesheets, selectors, declarations, properties, values, comments, the cascade, inheritance, and specificity.
Visual styling	Typography, colors, backgrounds, borders, spacing, link states, readable contrast, and consistent visual identity.
Box model	Content, padding, border, margin, width, max-width, min-height, box-sizing, spacing systems, and layout debugging.
Responsive design	Viewport meta tag, relative units, max-width, flexible images, breakpoints, media queries, and mobile-first thinking.
Flexbox layout	One-dimensional layout for navigation bars, cards, horizontal/vertical alignment, wrapping, spacing, and responsive reordering when appropriate.
CSS Grid layout	Two-dimensional page sections, card galleries, schedule/content grids, named or repeated columns, grid gaps, and responsive grid changes.
Testing and validation	Use browser developer tools, responsive design mode, CSS validation tools, and browser testing to identify and fix layout problems.

3. Learning Objectives

- Connect an external CSS file to multiple HTML pages and organize CSS rules clearly.
- Apply CSS selectors, classes, and reusable styling rules without relying on inline CSS.
- Use the CSS box model to control spacing, dimensions, borders, and visual separation.
- Create responsive pages that remain readable on desktop, tablet, and mobile screen widths.
- Use Flexbox to design navigation areas, card rows, and aligned interface elements.
- Use CSS Grid to create a structured two-dimensional page layout or content gallery.
- Use browser developer tools to inspect applied CSS, box model values, and responsive behavior.
- Validate CSS and prepare a short technical report explaining design decisions, testing results, and corrections.

4. Problem Description

Each team must design and implement a visually polished and responsive website for a fictional university event, club, workshop, technical mini-course, or academic service. Teams may continue the topic used in Laboratory 1, but the emphasis in this laboratory is different: the grading focus is now on CSS design, layout, responsiveness, and testing.

The website must use valid HTML5 and an external CSS file. The HTML structure should remain semantic and accessible, but the main objective is to demonstrate clear CSS styling, proper use of the box model, responsive design techniques, Flexbox, and CSS Grid.

5. Required Website Structure

File	Required Content
index.html	Home page introducing the website topic. Must include a responsive header, navigation, hero or introduction section, and at least one visually styled content area.
about.html	Information page explaining the event, club, workshop, or topic. Must include styled headings, images or visual content blocks, lists, and readable spacing.
schedule.html	Schedule, program, services, or activity page. Must include structured content presented using CSS Grid or a responsive layout, and may also include a styled table when appropriate.
contact.html or registration.html	Contact or registration page. Must include a styled and accessible form with responsive layout, visible focus styles, labels, and appropriate spacing.
styles.css	One external stylesheet used consistently across all pages. Additional CSS files may be used only if the structure is clearly explained in the report.

All pages must include consistent navigation and must be connected using relative hyperlinks. Each page must include the viewport meta tag and must be usable at different screen sizes.

6. Part A - CSS Organization and External Stylesheet

All CSS must be placed in one or more external stylesheet files. Inline CSS should not be used, except for rare cases that are explicitly justified in the report. The CSS file should be organized with comments and logical sections such as global styles, typography, navigation, layout, components, forms, and responsive rules.

- Use at least one shared external CSS file linked from every HTML page.
- Use class selectors for reusable components such as cards, buttons, containers, and grids.
- Avoid unnecessary repetition by creating reusable CSS rules.
- Use consistent naming for CSS classes.
- Include comments to identify the major sections of the CSS file.

```
<link rel="stylesheet" href="css/styles.css">
```

7. Part B - Visual Styling and Readability

The website must have a coherent visual design. The goal is not to create a complex commercial website, but the pages must be clean, readable, and visually organized.

- Use a consistent typography system for headings, body text, navigation, and buttons.
- Use a consistent color palette with sufficient contrast between text and background.
- Style navigation links, buttons, cards, tables, and form controls consistently.
- Use hover and focus states for interactive elements.
- Use spacing and alignment to make the content easy to scan.
- Ensure that images scale correctly and do not overflow their containers.

8. Part C - CSS Box Model

Students must demonstrate clear control of the CSS box model. The design should show intentional use of content width, padding, borders, margins, and box-sizing.

- Use box-sizing: border-box for predictable layout sizing.
- Use padding and margin to create readable spacing between sections and components.
- Use max-width and centered containers to prevent very long lines of text on large screens.
- Use borders, border-radius, or shadows where appropriate to visually separate components.
- Debug and explain at least one box model issue using browser developer tools in the report.

```
* {
  box-sizing: border-box;
}

.container {
  width: min(100% - 2rem, 1100px);
  margin: 0 auto;
}
```

9. Part D - Responsive Design and Media Queries

The website must respond to different screen sizes. Teams must test the website using browser responsive design tools and provide screenshots showing desktop, tablet, and mobile views.

- Include <meta name="viewport" content="width=device-width, initial-scale=1.0"> on every HTML page.
- Use relative units such as %, rem, em, fr, vw, or min()/max()/clamp() where appropriate.
- Use media queries to adjust navigation, grids, spacing, and typography at smaller screen widths.
- Ensure that no important content is cut off horizontally on mobile screens.
- Ensure that text remains readable and controls remain easy to use on mobile screens.

```
@media (max-width: 700px) {
  .main-nav {
    flex-direction: column;
    align-items: stretch;
  }

  .card-grid {
    grid-template-columns: 1fr;
  }
}
```

10. Part E - Flexbox Layout

Students must use Flexbox for at least two meaningful layout sections. Flexbox is especially appropriate for navigation bars, horizontal card groups, aligned buttons, form groups, or footer content.

- Use display: flex in at least two different website sections.
- Use properties such as justify-content, align-items, gap, flex-wrap, flex-direction, or flex-basis.
- Use Flexbox to improve alignment and spacing, not only to satisfy the requirement.
- Explain in the report why Flexbox was chosen for each selected section.

```
.main-nav {
  display: flex;
  gap: 1rem;
  justify-content: center;
  align-items: center;
  flex-wrap: wrap;
}
```

11. Part F - CSS Grid Layout

Students must use CSS Grid for at least one meaningful two-dimensional layout. Examples include a responsive card gallery, a schedule grid, a services grid, a team member grid, or a page section with main content and sidebar.

- Use display: grid in at least one major section of the website.
- Use grid-template-columns, gap, and responsive column changes.
- Use repeat(), minmax(), or fractional units where appropriate.
- Ensure that the grid remains readable on small screens.
- Explain in the report why CSS Grid was appropriate for that section.

```
.card-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 1.25rem;
}
```

12. Part G - Accessibility-Aware Styling

CSS must improve the user experience without reducing accessibility. Visual styling must not remove important cues used by keyboard users or users with visual difficulties.

- Do not remove focus outlines unless a clear replacement focus style is provided.
- Use sufficient color contrast for text, buttons, navigation links, and form labels.
- Do not communicate important information using color alone.
- Ensure that form controls remain readable and easy to identify.
- Ensure that responsive layout changes preserve logical reading order.

```
a:focus,
button:focus,
input:focus,
select:focus,
textarea:focus {
  outline: 3px solid #8f001a;
  outline-offset: 2px;
}
```

13. CSS and Responsive Design Checklist

Each team must complete the following checklist and include it in the report.

Item	Completed?
All pages link to the external stylesheet correctly.	

Inline CSS is avoided or explicitly justified.	
A consistent color and typography system is used.	
The CSS box model is applied intentionally using margin, padding, border, and box-sizing.	
At least two meaningful sections use Flexbox.	
At least one meaningful section uses CSS Grid.	
Images and content blocks are responsive and do not overflow.	
Media queries are used to improve the layout on smaller screens.	
The website was tested in desktop, tablet, and mobile widths.	
Interactive elements have visible hover and keyboard focus states.	
The CSS was checked using a validator or equivalent tool.	
Major CSS or layout errors were fixed.	

14. Laboratory Demonstration Requirement

Each team must demonstrate the laboratory work to the TA. The report will be considered for grading only if the demonstration has been completed. All team members must be present during the demonstration.

During the demonstration, the TA may ask any team member to explain the stylesheet organization, the visual design choices, the box model decisions, the Flexbox and Grid sections, the responsive breakpoints, the testing process, and the role of each submitted file. Each student must be able to explain the team submission clearly.

15. Testing Requirements

Each team must test the website before submission. The testing must include:

- Opening each HTML page in a modern browser.
- Checking that all internal navigation links work correctly.
- Checking that the external stylesheet loads correctly on every page.
- Testing the website at desktop, tablet, and mobile widths using browser responsive design mode.
- Inspecting at least one layout element using the browser developer tools box model panel.
- Testing the Flexbox and Grid sections at different widths.
- Checking that images and cards do not overflow their containers.
- Checking keyboard navigation and visible focus states.
- Validating the CSS using a CSS validator or equivalent tool and saving evidence of the results.

16. Deliverables

Each team must submit one ZIP file on Brightspace containing:

- All HTML files used for the website.
- All CSS files, including the main external stylesheet.
- Any image or media files used in the website.
- A short lab report in PDF format.
- Screenshots showing the website pages in a browser.
- Screenshots showing desktop, tablet, and mobile views.

- Screenshots or exported results showing CSS validation.
- Optional: a short README file explaining how to open and test the website.

17. Report Requirements

The lab report must include:

- Course code, lab number, team members, and student numbers.
- A short description of the website topic and the design goal.
- A file structure diagram or list explaining all submitted files.
- A description of the CSS organization and major selectors/classes used.
- A description of the visual design choices, including typography, colors, and spacing.
- An explanation of how the CSS box model was used in the website.
- An explanation of the Flexbox sections and why Flexbox was appropriate.
- An explanation of the CSS Grid section and why Grid was appropriate.
- A description of the responsive design approach, breakpoints, and media queries.
- The completed CSS and responsive design checklist.
- Screenshots of desktop, tablet, and mobile views.
- Screenshots of CSS validation results and any corrections made.
- A short reflection on problems encountered and how they were solved.

18. Grading Rubric

Component	Points
Correct multi-page website structure, file organization, and external stylesheet linking	10
CSS organization, reusable selectors/classes, and maintainable stylesheet structure	10
Visual styling: typography, colors, spacing, navigation, buttons, cards, and readability	15
Correct and intentional use of the CSS box model	15
Responsive design, media queries, flexible units, and mobile/tablet/desktop usability	15
Meaningful use of Flexbox in at least two layout sections	10
Meaningful use of CSS Grid in at least one major layout section	10
Accessibility-aware styling: contrast, focus states, readable forms, and keyboard-friendly design	5
Testing, CSS validation, responsive screenshots, and evidence of corrections	5
Laboratory demonstration to the TA with all team members present	3
Report quality, clarity, screenshots, and reflection	2
Total	100

19. Important Notes

- Late submissions will not be accepted unless an official accommodation or approved exceptional circumstance applies.
- All team members must understand the submitted website and report.

- External code, templates, images, fonts, icons, or resources must be cited if used and must be permitted for academic use.
- Students must not submit a website generated entirely by an external tool without understanding, adapting, and documenting the work.
- Academic integrity rules apply to all submitted files and reports.
- The use of CSS frameworks such as Bootstrap or Tailwind is not permitted for this laboratory unless explicitly approved by the instructor. The objective is to practice writing CSS manually.

20. Suggested Timeline

Date	Suggested Work
Monday, June 1	Read the lab statement, confirm the website topic, review the Lab 1 structure if reused, and plan the visual design.
June 2-3	Create or clean the HTML file structure and link the external stylesheet to all pages.
June 4-5	Implement the global CSS, typography, colors, navigation styling, and box model spacing.
June 6-7	Implement the Flexbox sections and CSS Grid layout sections.
June 8-9	Add responsive design rules and test desktop, tablet, and mobile views.
June 10	Use browser developer tools to inspect layout, fix overflow and spacing issues, and test keyboard focus states.
June 11	Validate the CSS, capture screenshots, and fix major validation or layout errors.
June 12	Complete the report, checklist, reflection, and final testing.
Saturday, June 13	Submit the ZIP file on Brightspace by 11:59 PM.